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Where the neurologists are not: measuring the gap between neurological disease burden and the capacity to treat it

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Neurological disease burden and the workforce trained to treat it are not distributed alike. Combining the WHO/WFN Neurology Atlas, the WHO Global Dementia Observatory and WHO Global Health Estimates across 52 countries, that mismatch is measured in three ways. First, a scale-free burden-to-capacity share ratio places the median low- or middle-income country at 13.1, meaning it carries thirteen times as large a share of the sample's stroke burden as of its neurologists, against 0.38 for the median high-income country: a factor of 34. Two countries report zero neurologists and cannot be placed on the scale. Second, capacity is distributed across countries with a Gini coefficient of 0.61 (95% bootstrap CI 0.43–0.73), the poorer half of countries holding 6% of the sampled workforce. Third, national density overstates access wherever specialists remain urban: an Urban Concentration Index falls from 0.95 in low-income countries to 0.52 in high-income ones, and the effective density facing a rural resident of a low-income country is zero, because no low-income country in the Atlas reports a neurologist practicing rurally. Since country-level regressions on 19 to 35 observations are underpowered, all inference is on binned groups. A bounding exercise shows burden would have to be overstated by a factor of 34 in low- and middle-income countries for the central result to reverse. All analysis is descriptive; no causal effect is estimated.

health workforce | neurological disease | global health equity | rural access | inequality measurement

Significance

Specialists per country is the statistic normally used to describe where medical capacity sits, and it answers the wrong question. Combining three World Health Organization sources across 52 countries, this study shows that the countries holding the largest share of the world's stroke burden hold the smallest share of its neurologists, by a factor of roughly thirty-four. It then shows that national counts overstate what most residents can reach, because the poorer a country is, the more completely its specialists concentrate in the capital. In low-income countries the specialist density facing a rural resident is not merely low but zero, which makes the usual language of ratios inapplicable.

Neurological disorders are among the largest contributors to global disability (1); the workforce to diagnose and manage them is among the most unevenly distributed of any medical specialty. Demand is rising fastest where capacity is thinnest: dementia prevalence may almost triple by 2050, with the steepest growth in low- and middle-income regions (2). Are countries carrying the heaviest neurological disease burden the ones with the capacity to treat it?

Related work. This is not the first analysis of neurological workforce scarcity. The WHO and World Federation of Neurology Atlas has documented that scarcity across two editions and established that resources are severely limited in low-income countries (3). Feigin and colleagues made a case for translating burden evidence into workforce policy (1), and comparative health-system statistics show specialist supply concentrated in high-income settings, with rural residents facing additional barriers even inside wealthy countries (4).

What the Atlas does not do, and what this analysis adds, is link capacity to need. The Atlas reports medians within income groups and WHO regions, presents no country-level workforce estimates, and includes no burden measure at all. It can therefore say that low-income countries have fewer neurologists, but not whether the countries with the least capacity are the ones carrying the heaviest burden, which is the question that matters for priority-setting. Grouped medians also suppress within-group variation, so a low-income country with functioning services and one with none are indistinguishable in the published tables. This extends that literature in three ways: it joins capacity to a burden measure at the country level, it reports a distributional statistic rather than a central tendency, and it separates how much capacity exists from where inside a country that capacity sits.

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A.P. designed the study, wrote all analysis code, performed the analysis, and wrote the paper.

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Data

0.0.0.1. Sources. Three WHO products and one survey inventory are used. (i) The *Global Dementia Observatory* (5) indicator `GDO_q6x1_2` gives neurologists per 100,000 population, self-reported by ministries of health. (ii) GDO indicator `GDO_q8x3_1` gives the furthest geographic level at which dementia diagnostic services are available, on an ordered three-level scale. (iii) WHO *Global Health Estimates* series `SA_0000001689` (6) gives age-standardized stroke DALYs per 100,000. (iv) The WHO/WFN *Neurology Atlas*, second edition (3), supplies the country to WHO-region to World-Bank-income-group crosswalk from Annex 1 and the aggregate workforce gradients from Tables 2–3 and Figures 11, 12 and 16. A DHS survey inventory is used only to describe coverage for a possible subnational extension.

0.0.0.2. Time window. Capacity and service reach are 2017, the fielding year of the second Atlas and the GDO round used. Burden is 2004, which is the only round of the WHO stroke DALY series exposed by the GHO API at the time of writing. The thirteen-year offset is the single largest weakness in the analysis and is the reason RQ1 is framed as describing a gap rather than estimating one.

0.0.0.3. Unit of analysis. In the original sources, the unit differs by product: GDO records are one row per country per indicator per round; GHE records are one row per country per cause per age-standardization per year; the Atlas Annex is one row per responding Member State. The unit of analysis used here is the country. Every source is collapsed to one row per ISO3 code before merging, and no country appears twice.

0.0.0.4. Missingness in key fields. The panel has 52 countries. Neurologist density is missing for 33 of 52 (63%), service reach for 5 (10%), income group for 13 (25%), and stroke burden for none. WHO encodes absence as the strings "Not available", "Not applicable", "No data" and "Not reported" rather than as empty cells; these are converted to missing in `code/03_clean_merge.py` (lines 19–20) before any numeric coercion, so that a placeholder string cannot become a zero. Missingness is not random: countries that report a workforce count plausibly have stronger health information systems, which biases the sample toward better-resourced countries and therefore understates the true gap.

0.0.0.5. Limitations of the sources. All WHO instruments here are ministry questionnaires, self-reported and unaudited, with no full-time-equivalent adjustment, so a neurologist working one clinic day a week counts the same as one working five. Stroke stands in for neurological burden generally. Measured burden itself depends on diagnostic capacity, which is the exposure of interest, so the two are connected inadvertently; this is addressed by bounding rather than by adjustment.

Methods

0.0.0.6. Cleaning. `code/03_clean_merge.py` performs every cleaning step. WHO placeholder strings are mapped to missing (line 20); the ordered service-reach question is mapped from text to an integer scale via `ACC_SCALE` in `utils.py` (line 36); and two binary indicators are derived from that scale, `beyond_capital` for levels ≥ 2 and `reaches_rural` for level 3 (lines 42–43).

0.0.0.7. Merging. Capacity and service reach are combined with an **outer join on ISO3**, deliberately, so that a country reporting one but not the other is retained, and is not dropped; this yields 52 rows. Burden and the Atlas crosswalk are then attached with **left joins on ISO3**, preserving those 52 rows. All matching is **exact on ISO3 country code**, is precise on country name, which avoids the standard failure mode where "Republic of Korea" and "Korea, Rep." fail to match. The script prints row counts before and after every join, and the printed counts appear in `output/log_03_clean_merge.txt`.

0.0.0.8. Variable definitions. The **burden-to-capacity share ratio** for country i is

$$R_i = \frac{b_i / \sum_j b_j}{c_i / \sum_j c_j} \quad [1]$$

where b_i is the age-standardized stroke DALY rate and c_i is neurologist density. Both terms are shares, so R_i is invariant to the units of either input. $R_i = 1$ means country i holds exactly as large a share of the sample's burden as of its workforce; $R_i = 13.1$ means the same staff is stretched over thirteen times the need. It is undefined at $c_i = 0$, and those countries are reported separately rather than dropped.

The **Urban Concentration Index** and **effective rural density** are

$$\mathrm{UCI}_g = \frac{p_g^{\mathrm{cap}} - p_g^{\mathrm{rur}}}{100} \quad [2]$$

$$D_g^{\mathrm{eff}} = m_g \cdot \frac{p_g^{\mathrm{rur}}}{100} \quad [3]$$

where p_g^{cap} and p_g^{rur} are the percentages of countries in income group g reporting neurologists in the capital and in rural areas, and m_g is the group's median density. UCI is bounded $[0, 1]$; a value of 1 means every country has capital coverage and none has rural coverage. D_g^{eff} rescales national density by the share of countries where a rural resident could reach anyone at all.

Inequality is measured by the **Gini coefficient** of neurologist density, computed by the sorted-index formula in `utils.py` and independently by trapezoidal integration under the Lorenz curve (line 186). The two agree to 1.1×10^{-16}, which is a check on the implementation.

0.0.0.9. Inference, and why no country-level regression is reported. This is a descriptive analysis; there is no source of causal identification: no randomization, no instrument, no policy discontinuity, and national income drives both capacity and measured burden. No causal effect is estimated or implied.

With 19 to 35 observations, a fitted country-level slope would be badly underpowered, so no slope is reported as a main result. Countries are binned by World Bank income group and the groups compared directly, with a Welch t-test, a Mann-Whitney U, a 20,000-draw permutation test that assumes nothing about distributional shape, a one-way ANOVA and Kruskal-Wallis across three bins, a chi-square on the reach-by-income contingency table, and a logistic regression of rural reach on high income. A regression is fitted in `code/07_regression_and_interactions.py`, but only to demonstrate two solvers agreeing (normal equations against gradient descent, to 4.4×10^{-16}) and to test an interaction the sample cannot resolve.

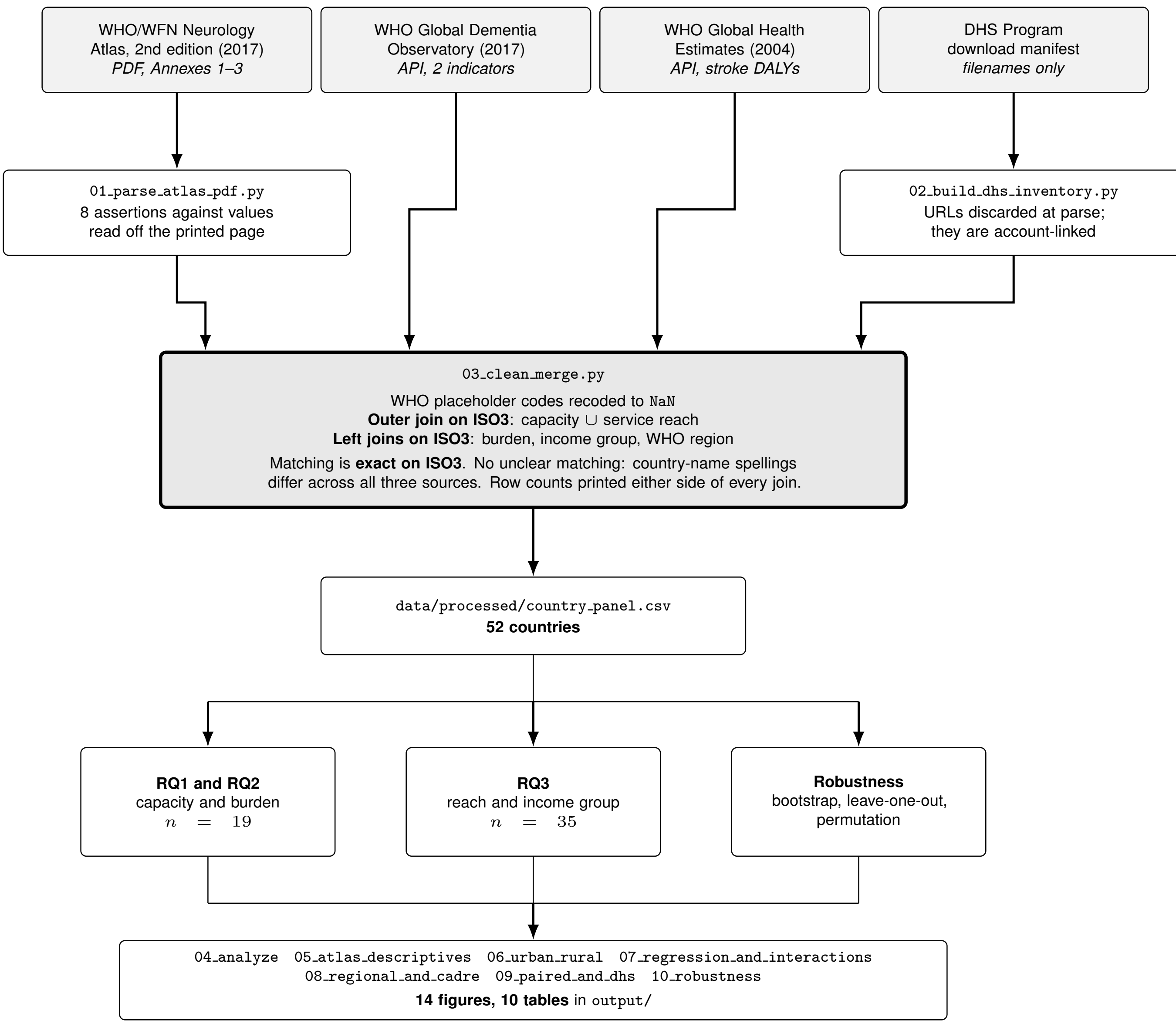


Fig. 1. The analysis pipeline, in the order it runs. Grey boxes are sources taken as given; the large box is the only step at which records from different sources are combined. The three analysis samples differ in size because reporting a workforce count and reporting a service-reach level are separate questionnaire items, answered by overlapping but non-identical sets of countries.

Results

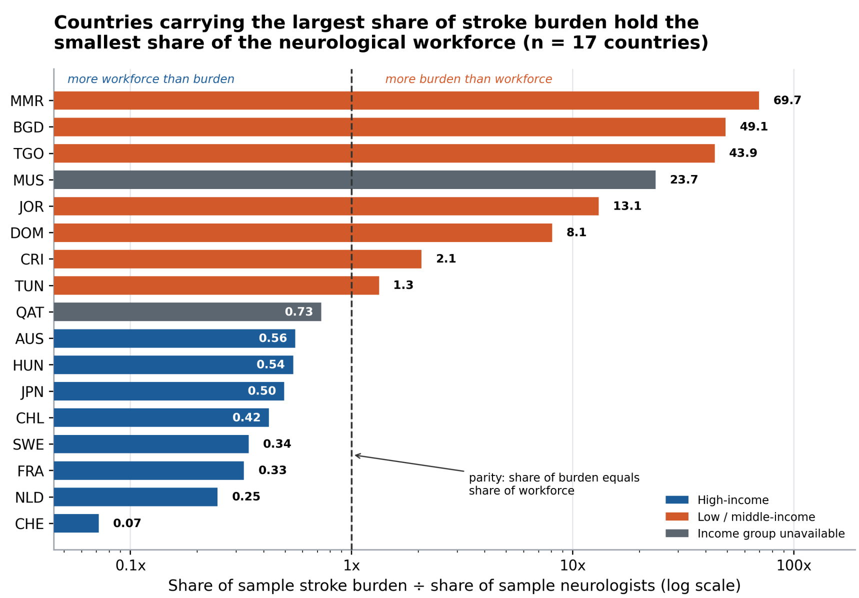
0.0.0.10. The analysis sample. Table 1 characterizes it. Of 52 countries in the merged panel, 19 support the RQ1 comparison and 35 the RQ3 comparison; the difference is entirely driven by which WHO indicators a country chose to answer.

Table 1. The analysis sample: row counts are after the merges described in Methods; every source is 2017 except stroke burden, which is 2004. The two analysis samples differ because reporting a workforce count and reporting a service-reach level are separate questionnaire items, answered by overlapping but non-identical sets of countries.

Measure	n
Countries in merged panel	52
with neurologist density (WHO GDO, 2017)	19
with service-reach level (WHO GDO, 2017)	47
with stroke DALY rate (WHO GHE, 2004)	52
with World Bank income group (Atlas, 2017)	39
RQ1 analysis sample: capacity and burden	19
RQ3 analysis sample: reach and income group	35

Countries reporting zero neurologists are retained in the panel but cannot be placed on the ratio scale of Eq. 1.

Capacity is not aligned with burden. On the share ratio, the median high-income country sits at 0.38 and the median low- or middle-income country at 13.12, a factor of 34.3. Myanmar reaches 69.7, Bangladesh 49.1, Togo 43.9; at the other end, Switzerland sits at 0.07. Eswatini and Fiji report zero neurologists and cannot be placed on the scale at all, while carrying stroke DALY rates of 1,536 and 994 per 100,000. Figure 2 shows the full ordering.

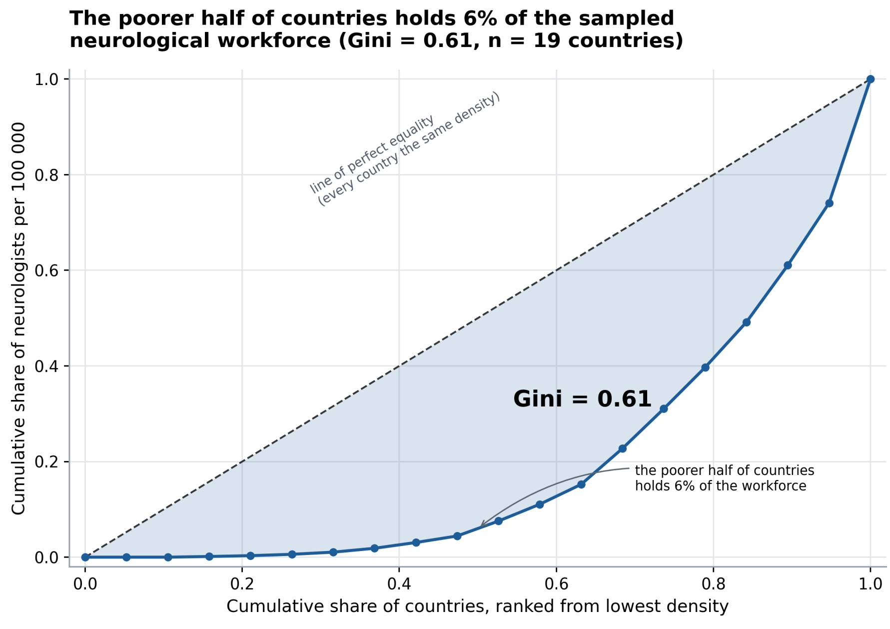


A ratio above 1.0 means the country holds a larger share of the sample's stroke burden than of its neurologists; below 1.0 the reverse. The axis is logarithmic because the ratio spans three orders of magnitude, which also places parity at the center. Capacity is WHO Global Dementia Observatory neurologist density per 100 000 (2017); burden is WHO Global Health Estimates age-standardized stroke DALYs per 100 000 (2004). Eswatini and Fiji report zero neurologists and cannot be placed on a ratio scale at all; their stroke DALY rates are 1536 and 994 per 100 000.

Fig. 2. Burden-to-capacity share ratio by country. The gap is a property of the whole distribution rather than of a few outliers: the two income groups barely overlap.

Capacity is distributed very unequally. The Gini coefficient of neurologist density across the 19 reporting countries is 0.61, with a 95% bootstrap interval of [0.43, 0.73]: high inequality, imprecisely estimated. The poorer half of countries holds about 6% of the sampled workforce. Inequality is greater between income groups than within them, with a Gini of 0.27 within high-income countries and 0.49 within low- and middle-income countries. The focal contrast is Switzerland at 13.10 per 100,000 against Myanmar at 0.07, an 187-fold difference. Dropping any single country moves the Gini by at most 0.06, so no one reporter carries the result.

The gap differs systematically between country types. Table 2 reports the binned comparisons; neurologist density differs between income bins on every test applied, including the distribution-free permutation test. A crucial descriptive result is that dementia diagnostic services reach rural areas in 77% of high-income countries and in none of the four low or lower-middle income countries sampled.



Lorenz curve of WHO Global Dementia Observatory neurologist density per 100 000 population (2017). Countries are ranked from lowest to highest density along the horizontal axis; the vertical axis is the cumulative share of the sampled workforce they hold. A Gini of 0 would place the curve on the dashed diagonal, a Gini of 1 in the bottom-right corner. The shaded area is half the Gini by construction. Highest density: CHE at 13.1 per 100 000. Lowest non-zero: MMR at 0.07. Two further countries report exactly zero and sit on the flat segment at the left.

Fig. 3. Lorenz curve and Gini of neurologist density. Most of the inequality sits between income groups rather than within them.

Table 2. Binned hypothesis tests; country-level regressions are not fitted as headline results because 19 to 35 observations cannot support them, so countries are binned by World Bank income group and the groups compared directly. Rank-based and permutation tests are reported alongside the parametric ones because the chi-square has six of nine cells with expected counts below five.

Comparison	Test	Result
Density, high vs low/middle	Welch t	$t = 4.29$, $P = 0.0031$
Same	Mann-Whitney U	$P = 0.0003$
Same	Permutation (20,000)	$P = 0.0003$
Reach across three bins	One-way ANOVA	$F = 5.74$, $P = 0.0074$
Same	Kruskal-Wallis	$H = 9.44$, $P = 0.0089$
Reach $\times$ income bin	Chi-square	$\chi^2 = 11.81$, $P = 0.0188$
Rural reach $\sim$ high income	Logistic	OR $= 7.6$, $P = 0.0098$

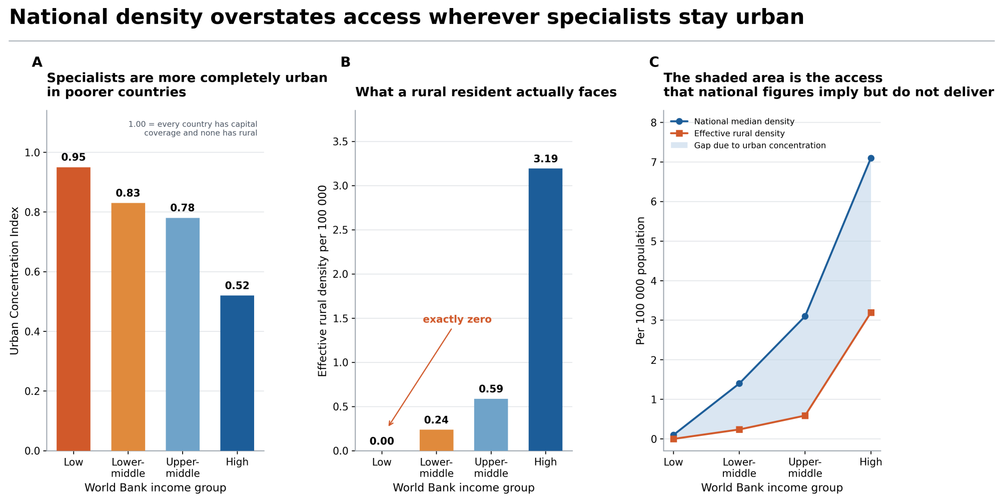
All tests are two-sided; the permutation test shuffles income-group labels 20,000 times and assumes nothing about distributional shape.

The gradient steepens with rarity. Across the 114 Atlas respondents, the high-to-low income ratio is 71-fold for total neurological workforce, 158-fold for adult neurologists, 62-fold for neurosurgeons and 195-fold for child neurologists; the scarcer the cadre, the more concentrated it is in rich countries.

National density overstates access. Figure 4 reports both indices. UCI falls monotonically from 0.95 in low-income countries through 0.83 and 0.78 to 0.52 in high-income ones. Effective rural density rises across the same groups from zero through 0.24 and 0.59 to 3.19 per 100,000. The zero is not a rounding artifact: no low-income country in the Atlas reports a single neurologist practicing rurally. The national gap between high- and low-income countries is 71-fold; the rural gap has no denominator at all.

An interaction the sample cannot resolve. If capacity buys less alignment where services remain urban, capacity and rural reach should interact. Only 13 countries carry both measures; the interaction coefficient is -0.004 with $P = 0.99$. That is a statement about statistical power, not evidence that no interaction exists.

Robustness. Three checks: A 5,000-draw bootstrap gives the Gini interval quoted above. Leave-one-out moves the Gini by at most 0.06 for any single country. A bounding exercise asks how much measurement error would be needed to overturn the central result: burden in low- and middle-income countries would have to be overstated by a factor of 34, or their capacity understated by the same factor, for the two median ratios to level. Measurement error of that magnitude is not plausible, and under-diagnosis in low-capacity settings pushes measured burden down rather than up, which widens the true gap rather than closing it.



WHO and World Federation of Neurology Atlas, 2nd edition (2017), Figures 12 and 16; 114 responding countries. Panel A: the Urban Concentration Index is the share of countries reporting specialists in the capital minus the share reporting any rural practice, divided by 100. It is bounded on [0, 1] because capital reach weakly dominates rural reach in every group observed. Panel B: effective rural density multiplies the national median by the share of countries with any rural practice, so it answers what a rural resident faces rather than what the national average is. It is exactly zero for low-income countries because no low-income country in the Atlas reports a neurologist practicing rurally, which makes the high-to-low gap undefined rather than merely large. Panel C plots both series on one axis.

Fig. 4. The two urban–rural indices. Panel B is the quantity a resident actually faces, and it reaches zero while the national measure in Panel A is still finite.

Discussion

Across three metrics, the finding is consistent: neurological capacity is concentrated where measured burden is lightest, and national counts overstate what most residents can reach. The policy implication is that specialists per country is the wrong statistic for priority-setting; it answers a question about supply when the operative question is about reach, and not for whether a rural resident can access anyone.

Several limitations bound the conclusion. The thirteen-year offset between capacity and burden is the weakest joint. The samples are small and self-selected toward countries with functioning health information systems, which understates the gap. Stroke stands in for neurological burden generally. Six of nine cells in the reach-by-income table have expected counts below five. Both WHO instruments are self-reported and unaudited with no FTE adjustment. The interaction test is underpowered at 13 countries.

Two extensions: The DHS inventory described in the repository covers 106 countries and 443 surveys, 62 with GPS coordinates, which would permit the same alignment question to be asked within countries rather than across them. A burden series contemporary with the 2017 capacity measures would remove the study's largest weakness.

Materials and Methods

All code, committed data and figures are at https://github.com/alannap27/qss20_neuro_access/tree/main, organized as numbered scripts in `code/`, committed inputs in `data/`, and figures and tables in `output/`. Shared functions are in `code/utils.py` and figure layout rules in `code/figstyle.py`. No script contains a hardcoded absolute path. The WHO inputs are public and committed; the Neurology Atlas PDF and the DHS manifest are excluded, the first because it is a copyrighted publication and the second because its URLs are tied to an approved DHS account. Analyses used Python 3.10 with pandas, NumPy, SciPy, statsmodels and Matplotlib; versions are pinned in `requirements.txt`. Random seeds are fixed throughout.

Data, Materials, and Software Availability. Every figure and table in this paper, and the eleven figures and eight tables not shown, are regenerated from the committed data by `bash run_all.sh`. The two scripts that rebuild `data/raw/` from restricted sources are excluded from that command and print an explanation rather than failing when their inputs are absent.

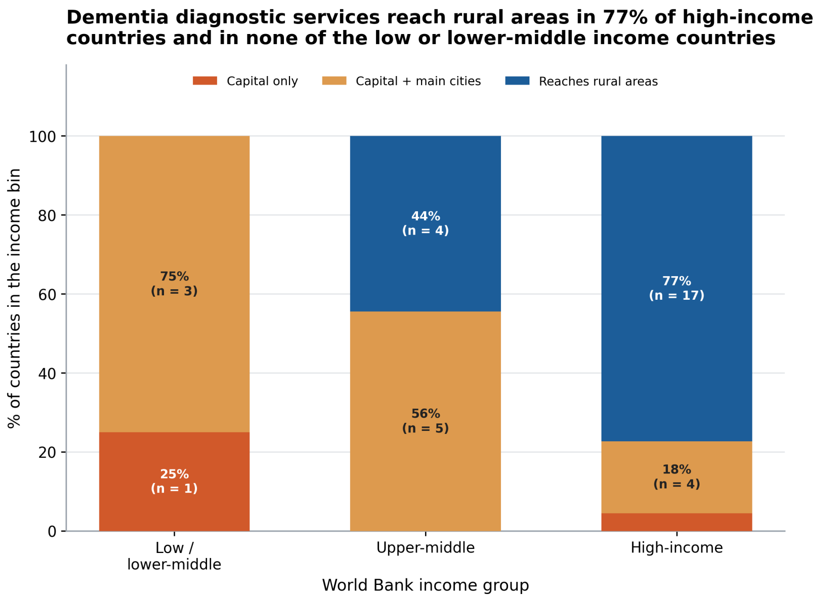
ACKNOWLEDGMENTS. I thank Prof. Herbert Chang for guidance on the analysis design and for the suggestion to bin countries rather than fit underpowered country-level regressions.

1. VL Feigin, T Vos, E Nichols, , et al., The global burden of neurological disorders: translating evidence into policy. *The Lancet Neurol.* **19**, 255–265 (2020).
2. GBD 2019 Dementia Forecasting Collaborators, Estimation of the global prevalence of dementia in 2019 and forecasted prevalence in 2050. *The Lancet Public Heal.* **7**, e105–e125 (2022).
3. World Health Organization, World Federation of Neurology, Atlas: Country resources for neurological disorders, 2nd edition, (World Health Organization, Geneva), Technical report (2017).
4. OECD, Health at a glance 2023: Oecd indicators, (OECD Publishing, Paris), Technical report (2023).
5. World Health Organization, Global dementia observatory (2017) Indicators GDO_q6x1_2 and GDO_q8x3_1, accessed via the WHO Global Health Observatory OData API.
6. World Health Organization, Global health estimates: age-standardized dalys, cerebrovascular disease (2004) Indicator SA_0000001689.

Appendix A. The remaining figures

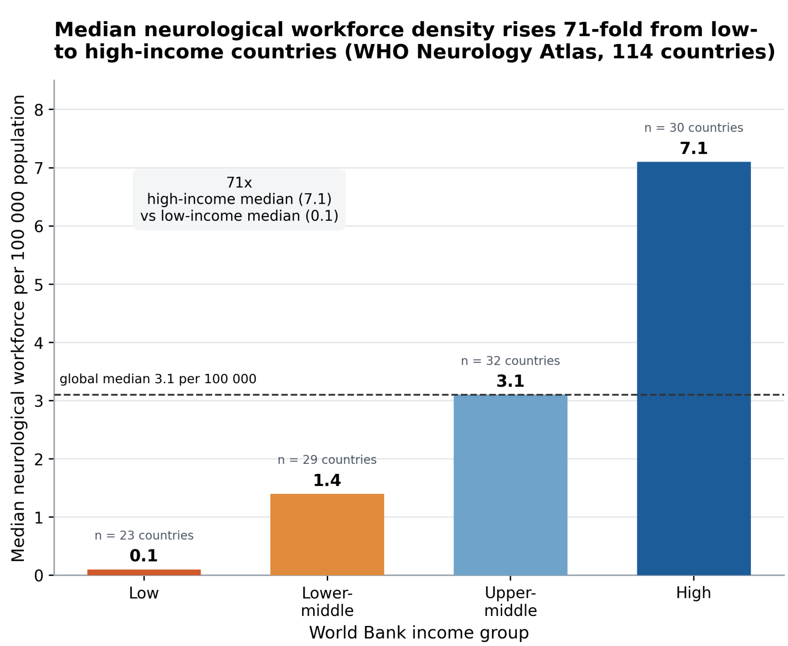
Three figures and two tables appear in the body; the pipeline produces fourteen figures and ten tables. The eleven not in the main paper are not weaker versions of those shown; they answer adjacent questions, and several exist to check the analysis rather than to say anything about neurologists. All are reproduced here with the same self-contained captions they carry in the repository.

A.1 Extending the descriptive picture.



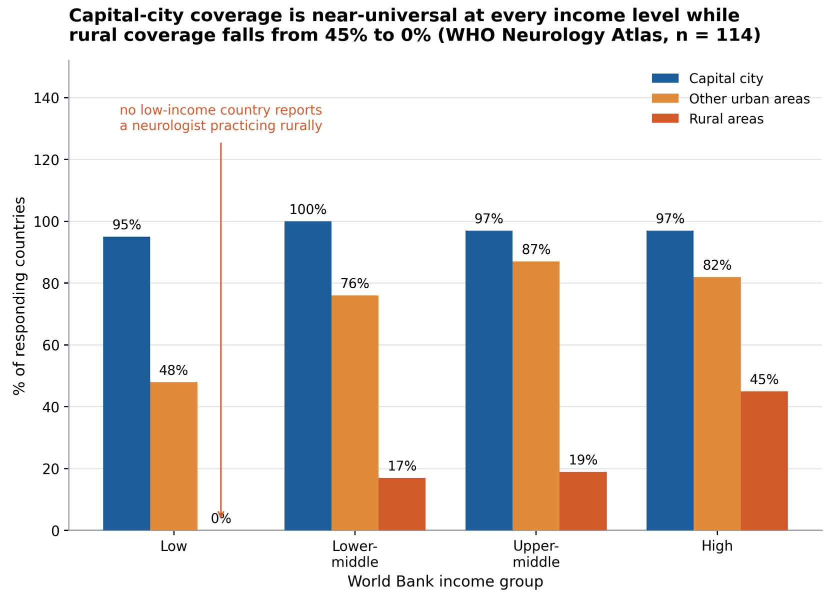
WHO Global Dementia Observatory (2017) indicator GDO_q8x3_1, cross-tabulated with World Bank income group taken from Annex 1 of the WHO Neurology Atlas. n = 35 countries reporting both fields. Bars sum to 100% within each income group because the question is single-choice. Chi-square = 11.81, df = 4, p = 0.0188, but 6 of 9 cells have an expected count below 5, so the rank-based and permutation tests reported in table t02 are the ones to rely on. Low and lower-middle income countries are pooled because only four low-income countries answered this item.

Fig. 5. Dementia diagnostic service reach by income group: the ordered outcome behind the binned tests in Table 2.



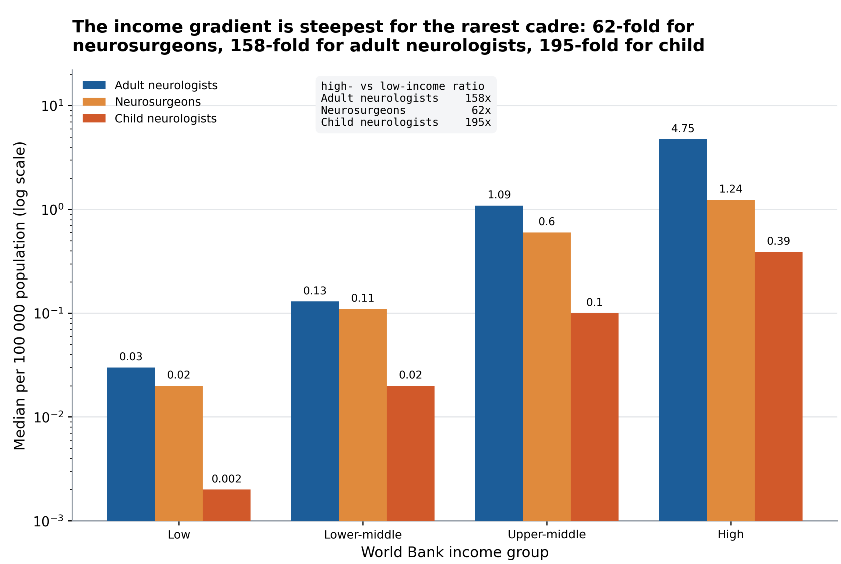
Median of the total neurological workforce, defined by the Atlas as adult neurologists plus neurosurgeons plus child neurologists, per 100 000 population within each World Bank income group. Source: WHO and World Federation of Neurology, Atlas: Country Resources for Neurological Disorders, 2nd edition (2017), Figure 12. The count under each bar is the number of countries answering that item. These are medians of country values and are not weighted by population, so a group median describes the typical country in the group rather than the experience of the typical person in it.

Fig. 6. Neurological workforce by income group across all Atlas respondents. The aggregate check on Fig. 2, on six times as many countries.



Share of responding countries reporting that neurologists practice in each setting, by World Bank income group. A country can report more than one setting, so bars within a group do not sum to 100%. Source: WHO and World Federation of Neurology Atlas, 2nd edition (2017), Figure 16; 114 responding countries. This is the aggregate counterpart to Figure 3, which measures the same gradient country by country in the smaller Global Dementia Observatory extract. The two instruments use different questions and different country sets and produce the same ordering.

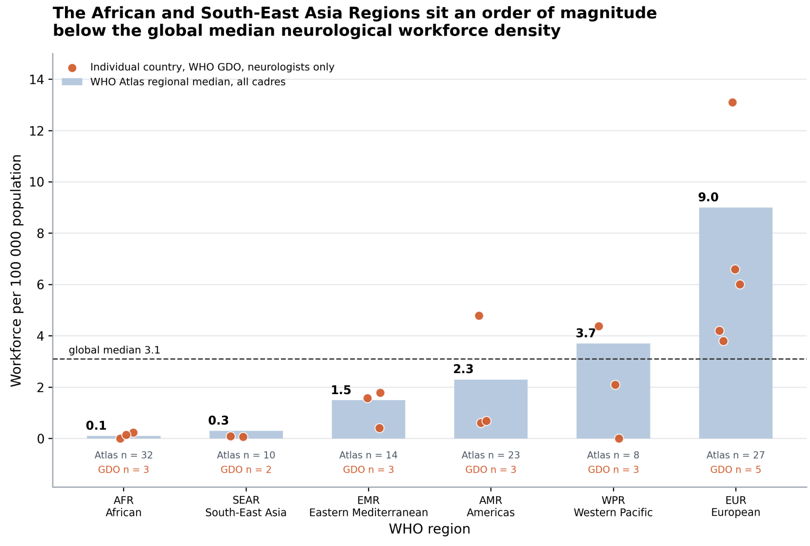
Fig. 7. Where neurologists practice, by income group. Capital coverage is near universal at every income level, so the entire gradient is in rural reach.



Median number of each cadre per 100 000 population by World Bank income group, on a logarithmic vertical axis because the values span four orders of magnitude, from 0.002 to 4.75. A logarithmic axis makes equal ratios equal distances, so the widening vertical spacing between the three series from left to right is itself the finding rather than an artifact of the scale. Source: WHO and World Federation of Neurology Atlas, 2nd edition (2017), Table 3. Response counts differ by cadre, from 93 countries for child neurologists to 114 for adult neurologists, and are reported in table t03.

Fig. 8. The income gradient by clinical cadre. Scarcity and concentration move together, which a single national count cannot show.

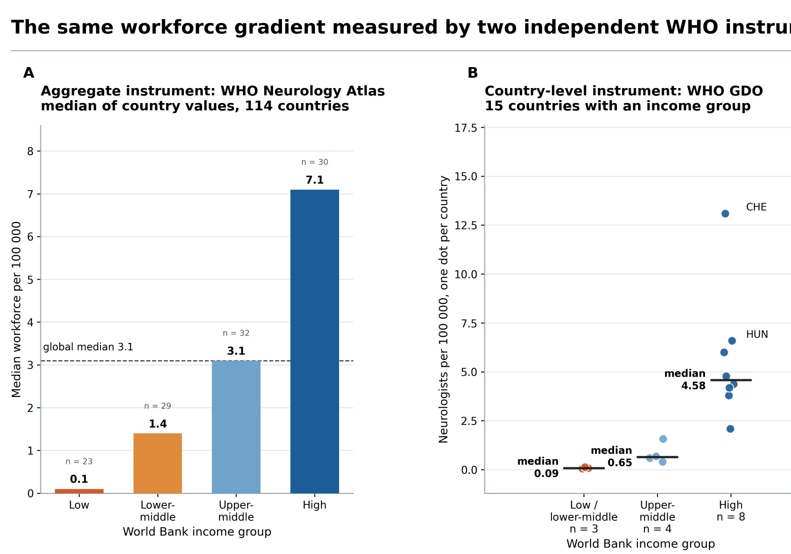
A.2 Geography rather than economics.



Bars are the WHO and World Federation of Neurology Atlas median of the total neurological workforce per 100 000 population within each WHO region (2017, Figure 11); the count under each bar is the number of countries answering that item. Orange dots are individual countries from the WHO Global Dementia Observatory (2017), which counts adult neurologists only and therefore sits systematically below the Atlas bar for the same region. Dots are jittered horizontally to avoid overplotting. Regions are ordered by Atlas median rather than alphabetically. The two instruments track each other closely: the GDO median for the European Region is 6.01 against the Atlas adult-neurologist median of 6.60, and for South-East Asia 0.08 against 0.10.

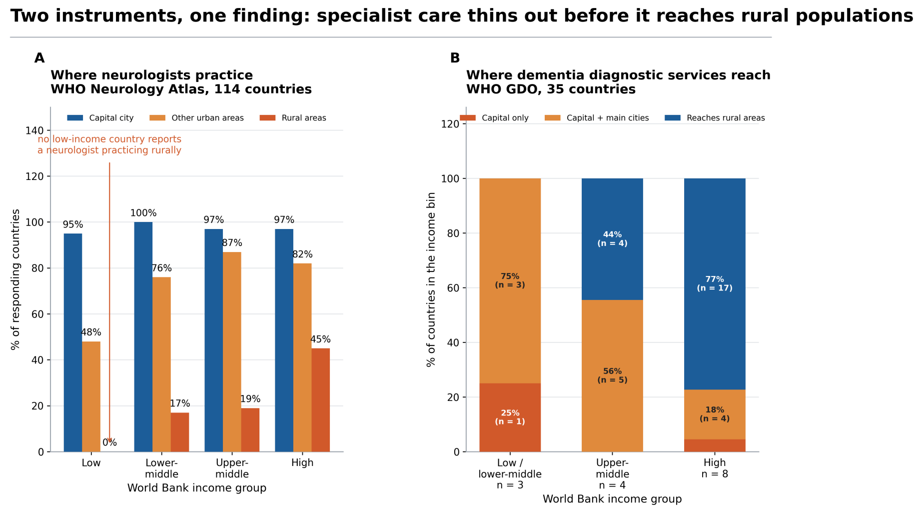
Fig. 9. Workforce density by WHO region; region and income group are correlated but not identical, and the country overlay is what separates them.

A.3 Paired instrument checks. Each panel places an Atlas aggregate beside its country-level counterpart from the Global Dementia Observatory. Two instruments collected independently, by different teams, with different questions, and ordering countries the same way is weak evidence that neither is simply wrong. It is not independent confirmation, because both are ministry self-reports.



Panel A reports the median across countries within each income group and cannot identify individual countries. Panel B reports each country separately but covers far fewer of them. The two disagree on level, because Panel A counts adult neurologists, neurosurgeons and child neurologists while Panel B counts adult neurologists only, and agree on direction and steepness. That agreement is the point: the country-level finding is not an artifact of the small sample. The three lowest-density countries in Panel B, too close together to label in place, are MMR 0.07, BGD 0.09, TGO 0.14 per 100 000. Sources: WHO and World Federation of Neurology Atlas 2nd edition (2017) Figure 12; WHO Global Dementia Observatory (2017) GDO_q6x1_2.

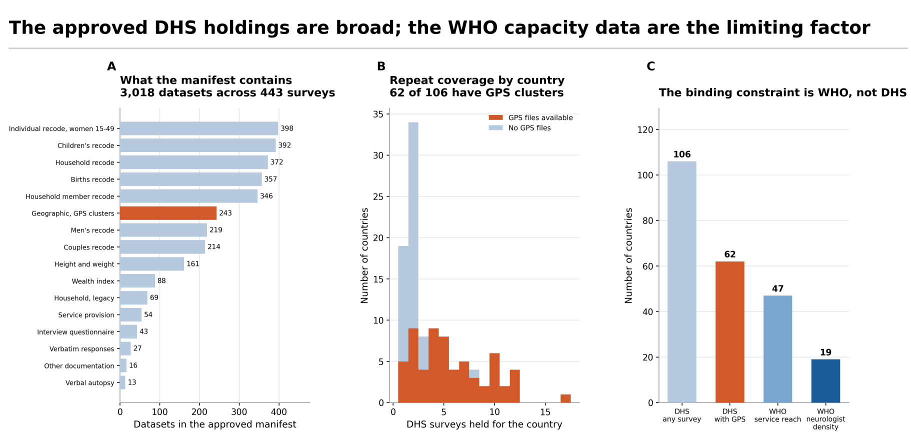
Fig. 10. The workforce gradient measured by two instruments. Agreement on direction across independent instruments is weak evidence that neither is simply wrong; both remain ministry self-reports.



Panel A asks whether neurologists practice in a given setting at all, and a country may report several. Panel B asks how far dementia diagnostic services extend, and the answer is single-choice. These are different questions about the same underlying constraint, asked of overlapping but not identical country sets, and they produce the same ordering: rural coverage falls from 45% to 0% across income groups in Panel A, and the share of countries whose services reach rural areas falls from 77% to 0% in Panel B. Sources: WHO Atlas 2nd edition (2017) Figure 16; WHO Global Dementia Observatory (2017) GDO_q8x3_1.

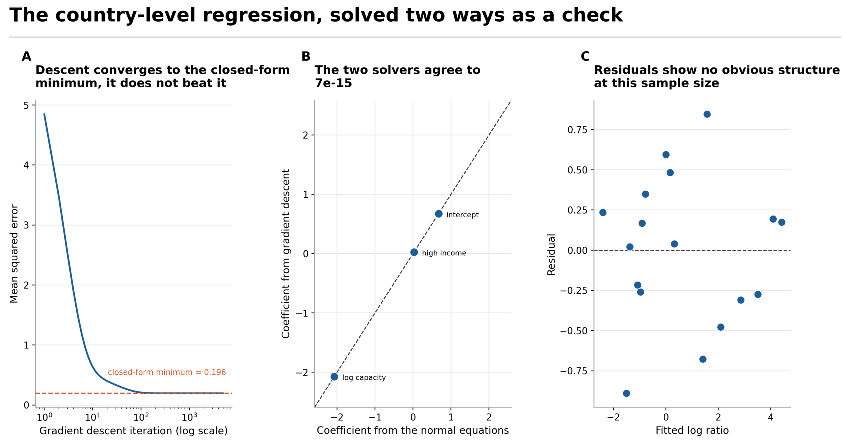
Fig. 11. The rural-access gradient measured by the same two instruments.

A.4 Groundwork and verification.



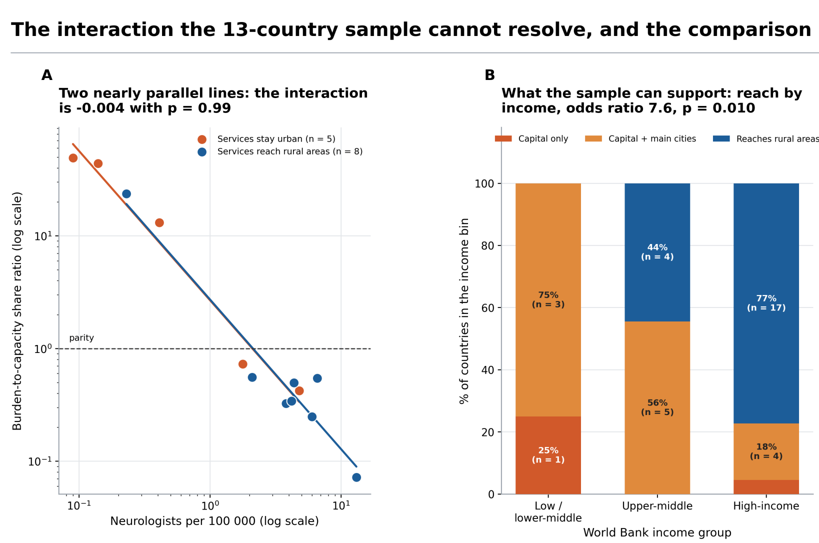
Derived from the filenames in the approved DHS Program download manifest. No survey data were downloaded or read: the DHS filename convention CCTTVVFL.zip encodes country, file type and survey phase, which is enough to build a coverage inventory. GE files carry displaced GPS cluster coordinates and are the input for the distance-to-care measure planned as the next step. Panel C is the reason the country-level analysis in Figures 1 to 3 is small: DHS covers 106 countries, but only 19 of them have a WHO neurologist-density value to link them to.

Fig. 12. DHS coverage across the sampled countries. Groundwork for asking the alignment question *within* countries rather than across them. It is not a result.



Outcome is the natural log of the burden-to-capacity share ratio for the 17 countries with both a burden and a positive capacity value. Predictors are standardized so a single learning rate suits both. Panel A: gradient descent minimizes the same mean squared error the normal equations solve exactly, so the curve approaches the dashed line from above and never crosses it. Panel B: coefficients from the two solvers plotted against each other; points on the diagonal mean they agree. Panel C: residuals against fitted values, the standard check for non-constant variance or curvature, neither of which is visible here, though 17 points is too few to see much. Capacity appears in the denominator of the outcome, so its large negative coefficient is mechanical rather than a substantive finding.

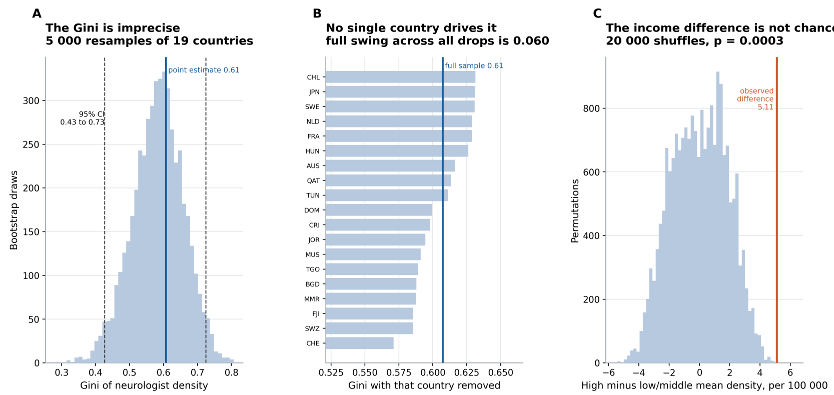
Fig. 13. The country-level regression solved two ways. This verifies the implementation and says nothing about neurologists.



Panel A: each point is one of the 13 countries carrying both a burden-to-capacity ratio and a service-reach level. Both axes are logarithmic. Lines are ordinary least squares fits within each group; if capacity bought less alignment where services stay urban, the orange line would be flatter than the blue one. It is not, and the interaction coefficient is indistinguishable from zero. That is a statement about statistical power at n = 13, not evidence that no interaction exists. Panel B shows the comparison the larger service-reach sample does support: rural reach is far more common in high-income countries. Source: WHO Global Dementia Observatory (2017) and WHO Global Health Estimates. Cell counts are small throughout, so both panels are descriptive.

Fig. 14. The interaction the 13-country sample cannot resolve, beside the comparison it can.

Robustness of the country-level results to their small sample



Panel A resamples the 19 countries with replacement 5 000 times and recomputes the Gini each time. The interval is wide because the sample is small, so the coefficient should be read as evidence of high inequality rather than as the precise value 0.61. Panel B recomputes the Gini dropping each country in turn, which checks that no single reporter carries the result; the full swing is 0.060. Panel C shuffles the income labels 20 000 times to build the null distribution for the difference in means directly, which avoids any normality assumption at n = 8 versus 7. All three use WHO Global Dementia Observatory neurologist density per 100 000 population (2017).

Fig. 15. Robustness of the country-level results to their small sample.

A.5 The remaining tables.

The eight tables not shown in the body are reproduced here; all are committed as CSV in `output/tables/` and are regenerated by `run_all.sh`.

Table 3. The income gradient by clinical cadre. Median per 100 000 population in low- and high-income countries, the absolute gap between them, and their ratio. Source: WHO/WFN Neurology Atlas, 2nd edition (2017), Table 3. This is the numeric backing for Fig. 8.

Cadre	Low	High	Gap	Ratio
Total neurological workforce	0.1	7.1	7	71
Adult neurologists	0.03	4.75	4.72	158.3
Neurosurgeons	0.02	1.24	1.22	62
Child neurologists	0.002	0.39	0.388	195

Table 4. The two urban–rural indices by income group. Density is total neurological workforce per 100 000; the three percentage columns are the share of countries reporting practice in each setting; UCI is the Urban Concentration Index; effective rural density is the national median rescaled by the share of countries with any rural practice. Numeric backing for Fig. 4.

Income group	n	Density	Capital	Urban	Rural	UCI	Eff. rural
Low-income	23	0.1	95%	48%	0%	0.95	0.000
Lower-middle-income	29	1.4	100%	76%	17%	0.83	0.238
Upper-middle-income	32	3.1	97%	87%	19%	0.78	0.589
High-income	30	7.1	97%	82%	45%	0.52	3.195

Table 5. The country-level regression solved two ways. Outcome is the natural log of the burden-to-capacity share ratio; predictors are standardized. The two solvers agree to 4.4×10^{-16}; numeric backing for Fig. 13.

Term	Normal eqs.	Grad. descent	SE	t
intercept	0.6747	0.6747	0.1185	5.695
log capacity (standardized)	-2.0715	-2.0715	0.2131	-9.723
high income (standardized)	0.0275	0.0275	0.2131	0.129

Table 6. The interaction tests; the capacity-by-rural-reach interaction is indistinguishable from zero at $n = 13$, which is a statement about power rather than evidence of no interaction. Numeric backing for Fig. 14.

Comparison	Test	Statistic	P
Rural reach on high income	Logistic regression	beta = +2.035	0.0098
log ratio on log capacity x reaches rural	OLS interaction	beta = -0.004	0.9911
Service reach across income bins	One-way ANOVA	F = 5.74	0.0074

Table 7. Workforce density by WHO region, per 100 000. The Atlas column counts three cadres and the Global Dementia Observatory column counts adult neurologists only, which is why the latter sits systematically lower. Numeric backing for Fig. 9.

Code	Region	Atlas	n	GDO med.	n
AFR	African Region	0.1	32	0.14	3
AMR	Region of the Americas	2.3	23	0.69	3
EMR	Eastern Mediterranean Region	1.5	14	1.58	3
EUR	European Region	9	27	6.01	5
SEAR	South-East Asia Region	0.3	10	0.08	2
WPR	Western Pacific Region	3.7	8	2.1	3
Global	Global	3.1	114	—	0

Table 8. The paired-instrument comparison. Each construct is measured by an Atlas aggregate and by a country-level Global Dementia Observatory extract; the two disagree on level and agree on direction. Numeric backing for Figs. 10 and 11.

Construct	Instrument	n	Finding
Workforce gradient	WHO Neurology Atlas, aggregate	114	0.1 to 7.1 per 100 000, 71-fold
Workforce gradient	WHO GDO, country level	15	median 0.09 to 4.58 per 100 000
Rural access	WHO Neurology Atlas, aggregate	114	45% to 0% across income groups
Rural access	WHO GDO, country level	35	77% to 0% of countries reach rural areas

Table 9. Demographic and Health Survey coverage, derived from filenames in the approved download manifest; no survey data were read. Numeric backing for Fig. 12.

Metric	Value
Datasets in manifest	3018
Distinct countries	106
Distinct surveys	443
Countries with GPS files	62
Countries with one survey only	19
Countries with five or more surveys	36
Median surveys per country	2.5

Table 10. Robustness of the country-level results. Numeric backing for Fig. 15.

Quantity	Estimate	Uncertainty
Gini of neurologist density	0.607	[0.427, 0.725]
Gini, leave-one-out range	0.571 to 0.631	swing 0.060
High- vs low/middle-income density	5.62 vs 0.51 per 100 000	permutation p = 0.0003
Confounding needed to erase the RQ1 gap	median ratio 13.12 vs 0.38	factor 34.3

Agentic Analysis

Transcripts are in `docs/agentic/`.

The research questions, the choice of a share ratio over a difference, the decision to bin rather than regress, and the bounding exercise are mine; I executed and read every line of code, and traced every reported number back to a source table. Claude was used for a few sparse coding errors throughout, as well as for a couple of questions whose answers were rejected. The space below is for the couple of scripts where the assistant was wrong (the times I had asked for help).

0.0.0.11. Rejected: a regression presented as a main finding. Claude proposed fitting the alignment ratio on capacity and income and reporting the slope as the paper's main quantitative result, with a standard error and a p-value.

How it was wrong: the estimate is not identified in any useful sense on 19 observations, but the output does not look that way. A coefficient with a standard error carries an implicit claim to precision that the sample cannot support, and nothing in the fitted object flags this. I replaced it with binned comparisons and a 20,000-draw permutation test, which make a weaker claim, but is more fully accurate. The regression survives in Fig. 13, as a check that two solvers agree.

0.0.0.12. Rejected: dropping the countries that break the plot. Eswatini and Fiji report zero neurologists, so the share ratio is undefined for them, and they cannot be drawn on a logarithmic axis; Claude proposed excluding them.

How it was wrong: the suggestion optimized for a clean figure over a correct sample. Those two countries are the most extreme instances of the phenomenon the paper is about, and their stroke DALY rates, 1,536 and 994 per 100,000, are among the highest in the sample. Dropping them would have removed the strongest evidence for the claim while improving the axis. They are reported separately in the text and named in the caption instead.

0.0.0.13. Failure: silent truncation in the PDF parser, and an over-correction. The Neurology Atlas wraps two long country names across lines. The first parser took only the first line, yielding "The former Yugoslav Republic of" as a country name.

How it was wrong, first attempt: the parse succeeded, produced 133 rows as expected, and the corrupted values sat in a column. Row counts are the usual check, and row counts were correct.

How it was wrong, second attempt: the assistant's fix appended any short following line to the country name. This over-corrected, gluing contributor names onto countries and producing entries such as "Austria Grisold, Regina Katzenschlager." The rule that held keys on the observation that a wrapped country name is cut mid-phrase and ends in a function word, which no contributor name does. Eight assertions against values read off the page now guard the parse.

0.0.0.14. What this implies. All the errors imply that the output is internally consistent and externally wrong, produced with the same confidence whether right or not. None would have been caught by a test suite, because in each case the code did what it said. Each surfaced only when I asked what a quantity meant rather than whether the script ran. AI assistance moved the bottleneck from writing code to verifying that the code answered the intended question, and the second is the harder job; thus, after these initial asks failed, barring a few errors (as mentioned above), AI assistance was minimized.

0.0.0.15. Website. AI was lastly used to develop the source code for the launched website https://vercel.com/alanna2/care-capacity-site with a descriptive prompt for the overview, most aspects were left blank to insert the graphs/information by hand.